

Quenched Bond Disorder in Non-Collinear Magnets

Santanu Dey

Technische Universität Dresden

Content

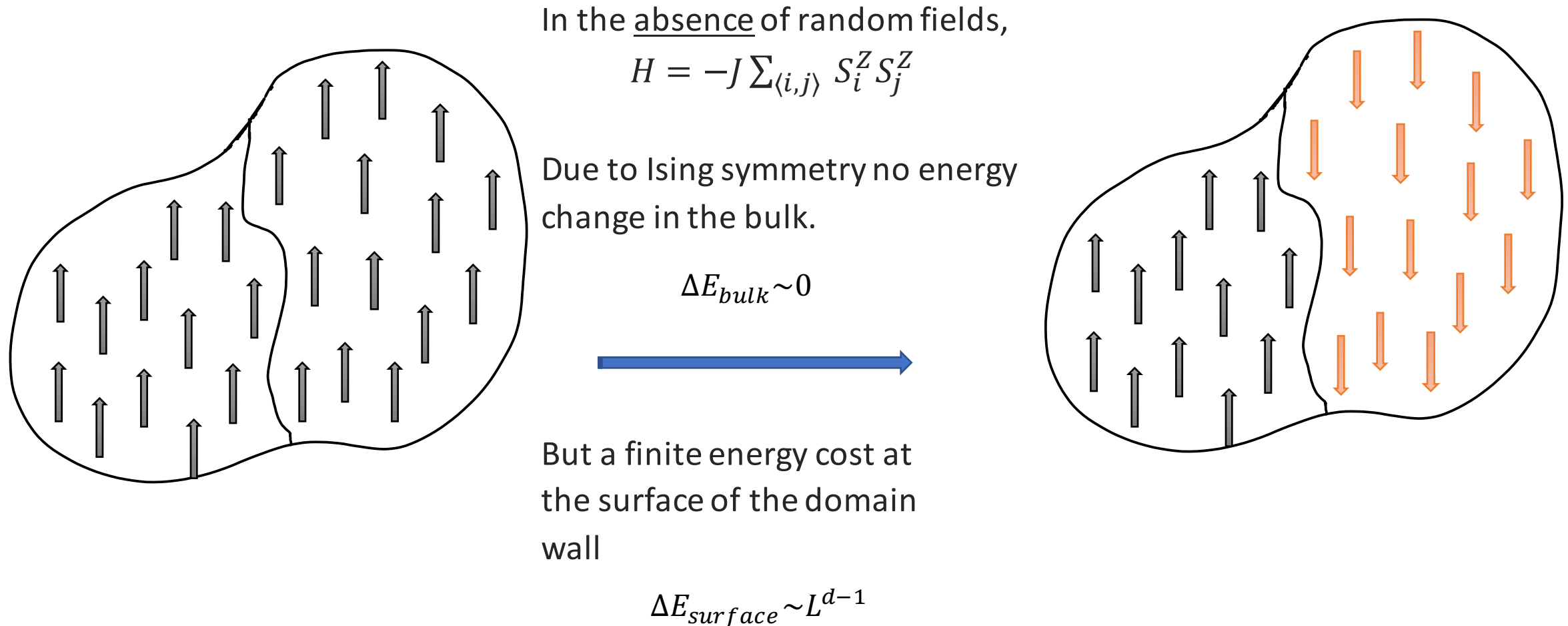
- Quenched disorder: Random fields
 - Random fields and the Imry-Ma criterion.
 - Separate lower critical dimension for discrete and continuous symmetry.
- Quenched disorder: Random couplings
 - Bond disorder in frustrated non-collinear magnets.
 - Rigid in-commensurability domains.
- An example
 - Disordered triangular lattice Heisenberg antiferromagnet.
 - Experiments on related systems

Quenched Disorder: Random Fields

Lower critical dimensions

Random fields and the Imry-Ma criterion

A simple picture with Ising spins:



Random fields and the Imry-Ma criterion

A simple picture with Ising spins:

In the presence of **random fields**,

$$H = -J \sum_{\langle i,j \rangle} S_i^Z S_j^Z + \sum_i \mathbf{h}_i S_i^Z$$

but this time the bulk energy is affected.

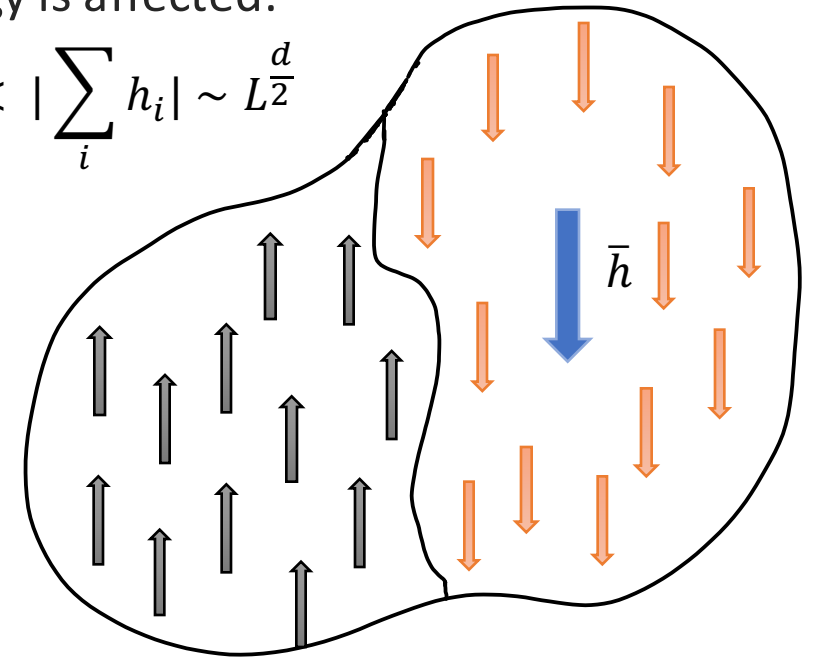
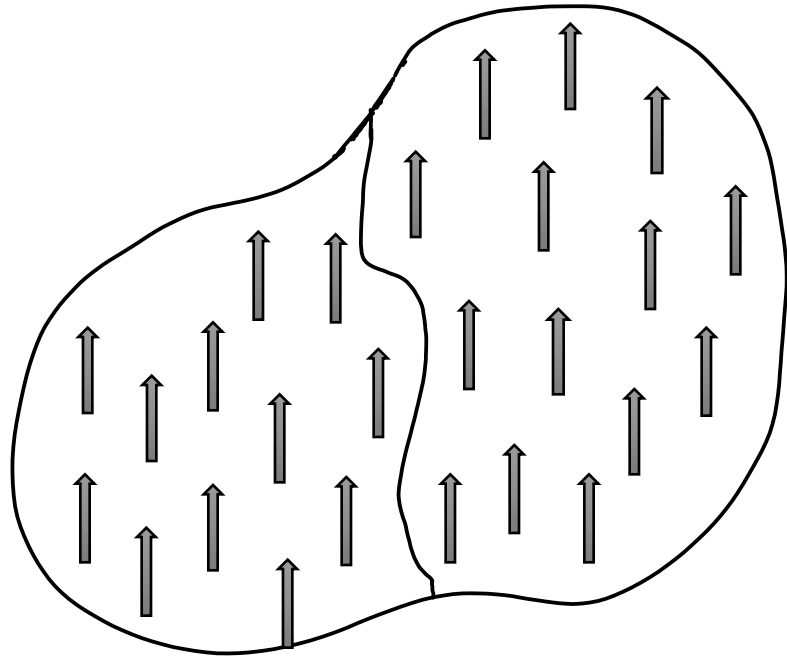
$$\Delta E_{bulk} \sim \sum_i h_i \Delta S_i^Z \sim \sum_i h_i < \left| \sum_i h_i \right| \sim L^{\frac{d}{2}}$$

A “typical value” for
possible energy gain.



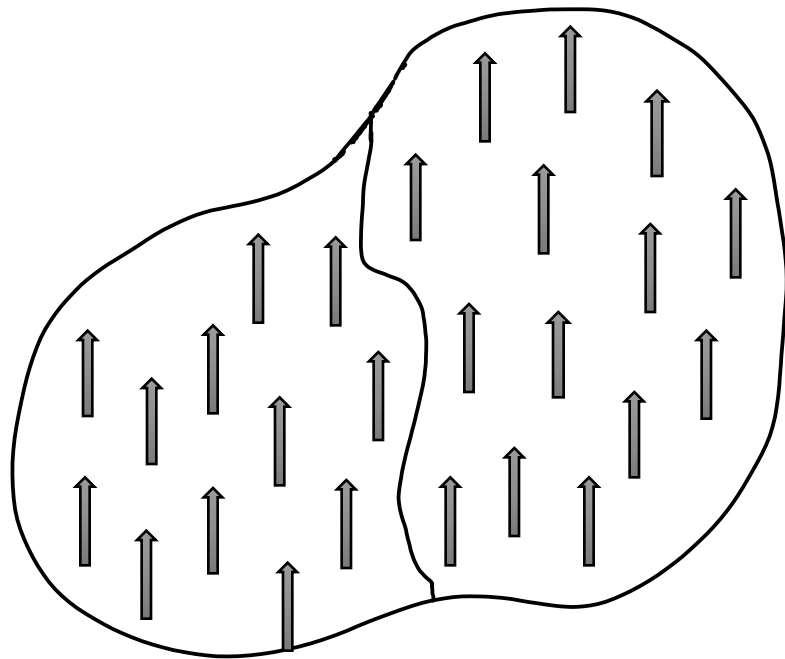
The energy cost is the same

$$\Delta E_{surface} \sim L^{d-1}$$



Random fields and the Imry-Ma criterion

A simple picture with Ising spins:



In the presence of **random fields**,

$$H = -J \sum_{\langle i,j \rangle} S_i^Z S_j^Z + \sum_i h_i S_i^Z$$

long range order is destroyed if,

$$\Delta E_{bulk} \geq \Delta E_{surface}$$

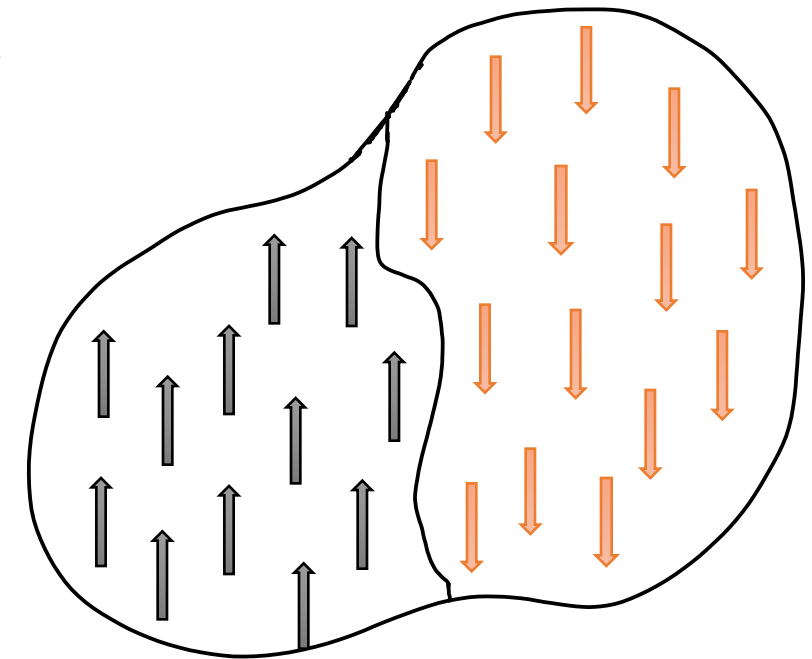
$$\Rightarrow L^{\frac{d}{2}} \geq L^{d-1}$$

$$\Rightarrow d \leq 2$$

This makes

$d_c = 2$ the lower

critical dimension for discrete symmetries.



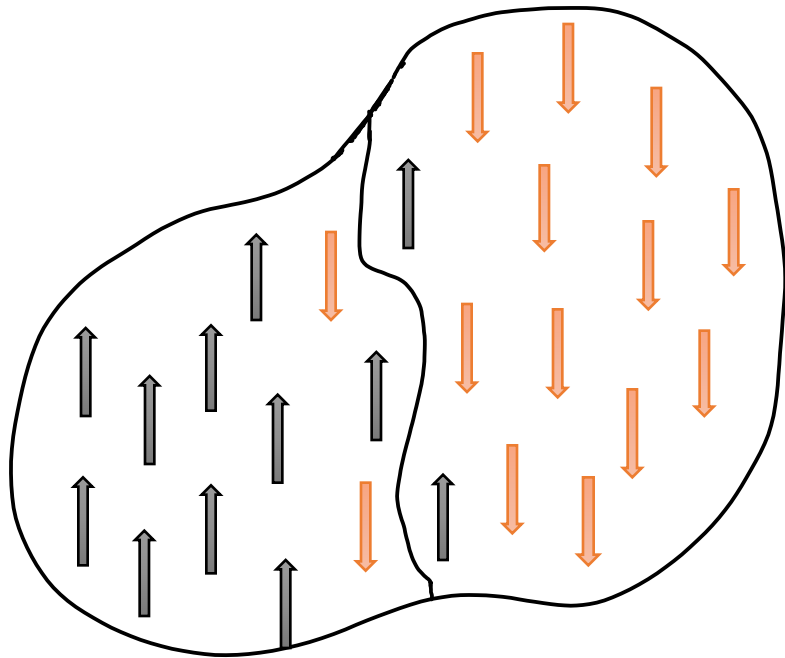
Favoured!

Y. Imry and S.-k. Ma, Phys. Rev. Lett. 35, 1399 (1975)

G. Grinstein and Shang-keng Ma, Phys. Rev. Lett. 49, 685 (1982)

Random fields and the Imry-Ma criterion

Rough domain walls:



- Domain walls can be rough and ill defined

- Domain wall thickness scales as,

$$w_d(L) \propto L^{\frac{5-d}{2}}$$

implying that for $L \leq 3$ domain walls thickness, $w_d(L) \gg L$, which, once included in the energy estimate gives a critical dimension,

$$d_c = 3 \quad [1,2]$$

- But an even more careful analysis shows that the scaling is in fact,

$$w_d(L) \propto L^{\frac{5-d}{3}}$$

- for $d > 2$, the ratio $\frac{w}{L} \rightarrow 0$, implying, $d_c = 2$ [3]

[1] E. Pytte, Y. Imry, and D. Mukamel, Phys. Rev. Lett. 46, 1173 (1981)

[2] K. Binder, Y. Imry, and E. Pytte, Phys. Rev. B 24, 6736(R) (1981)

[3] G. Grinstein and Shang-keng Ma, Phys. Rev. Lett. 49, 685 (1982)

Random fields and the Imry-Ma criterion

For continuous symmetries:

A **random field** Heisenberg model for illustration,

$$H = -J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j + \sum_i \vec{h}_i \cdot \vec{S}_i$$

Order parameter can smoothly vary between domains, costing less energy

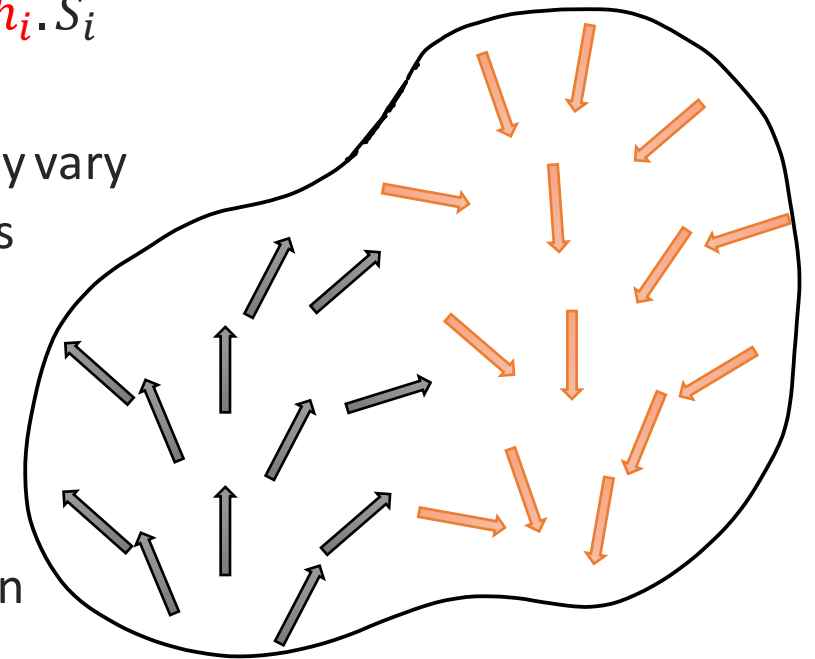
$$\Delta E_{bulk} \sim L^{d/2}$$

$$\Delta E_{surface} \sim L^{d-2}$$

So the lower critical dimension for continuous symmetries is,

$$d_c = 4$$

Y. Imry and S.-k. Ma, Phys. Rev. Lett. 35, 1399 (1975)



$$\Delta E_{surface} = \int d^d x \nabla \theta^2 \sim L^{d-2}$$

Random fields and the Imry-Ma criterion

Goldstone modes and infrared divergence:

- The order parameter field couples with the random fields, $\overline{h_r h_{r'}} = h_0^2 \delta_{rr'}$

$$\sum_r h_r \phi_r$$

- This leads order parameter fluctuation

$$\langle \phi_r \rangle = \sum_{r'} h_{r'} \langle \phi_r \phi_{r'} \rangle = \sum_{r'} h_{r'} \chi_{rr'}$$

$$\Rightarrow \overline{\langle \phi_r^2 \rangle} = h_0^2 \int \frac{d^d q}{(2\pi)^d} \chi(q)^2$$

$\int \frac{d^d q}{2\pi} \left(\frac{1}{q^2}\right)^2 \sim \text{infrared}$
divergent for $d \leq 4$

Goldstone modes, $\chi(q) \sim \frac{1}{q^2}$

Hence, the lower critical dimension for random field systems with continuous symmetry is,

Y. Imry and S.-k. Ma, Phys. Rev. Lett. 35, 1399 (1975)
A. Aharony, Solid State Comm., 28, 8, 667 (1978)

$$d_c = 4$$

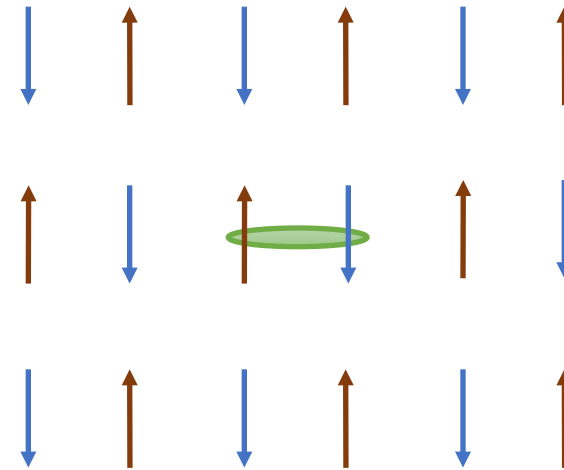
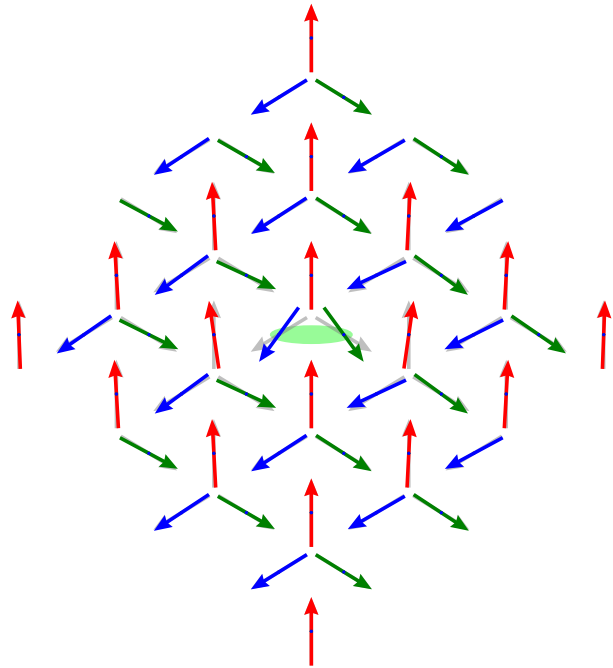
Quenched Disorder: Random Couplings

Lower critical dimension of disordered non-collinear magnets: Infrared divergence

Single bond impurity

Local reorientation for non-collinear magnets

- In a non-collinear Heisenberg antiferromagnet
- In a collinear Heisenberg antiferromagnet

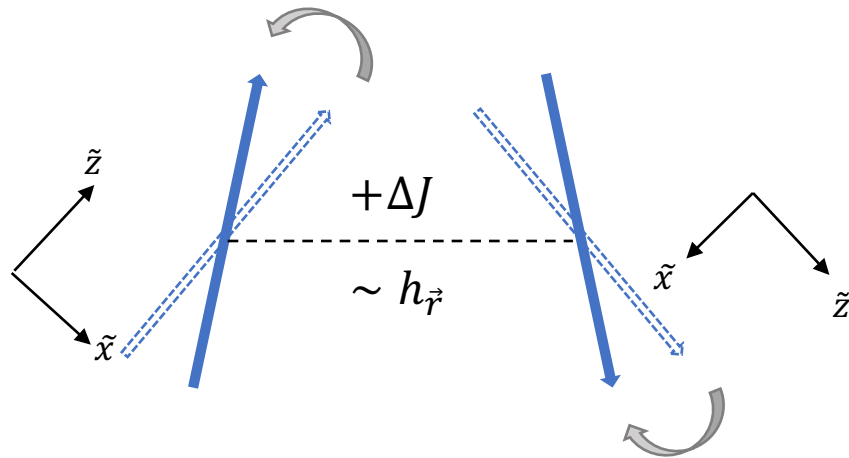


Bond disorder is weaker and it does not break time reversal invariance.

Single bond impurity

Local reorientation for non-collinear magnets

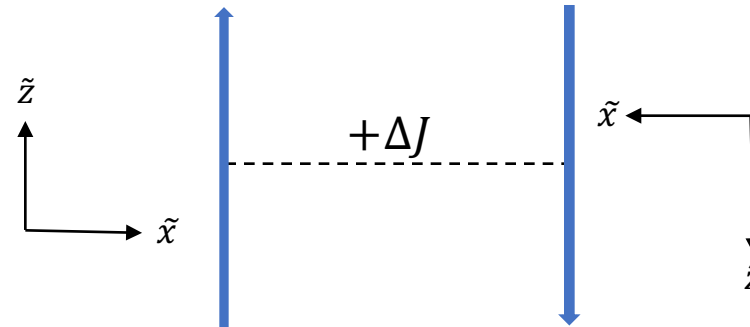
- In a non-collinear magnet



Frustration is locally relieved. For infinitesimal disorder, this is a perturbative correction to the “effective” bulk Hamiltonian,

$$H \simeq H_0 - d_{\vec{r}}(\tilde{S}_{\vec{r}}^x - \tilde{S}_{\vec{r}+\vec{\delta}}^x)$$

- In a collinear magnet



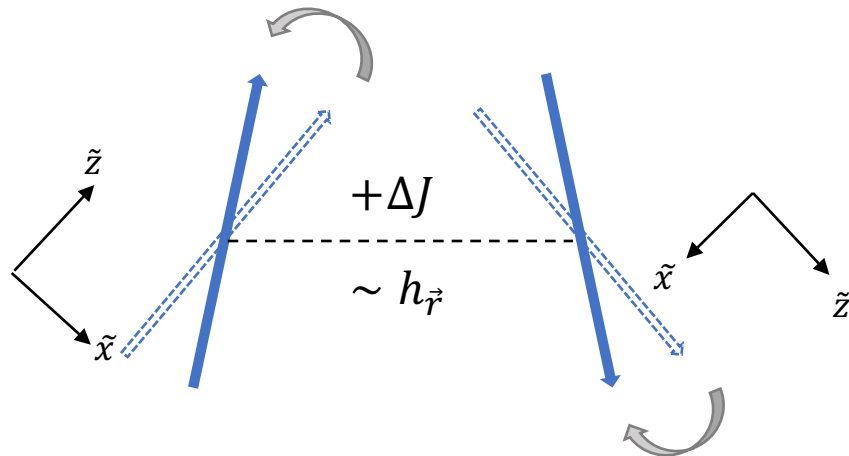
There is no change in the bulk order in response to the bond impurity, hence the Hamiltonian is “virtually” unchanged

$$H \simeq H_0$$

Single bond impurity

Dipolar response for the Goldstone modes

- In a non-collinear magnet



Frustration is locally relieved. For infinitesimal disorder, this is a perturbative correction to the “effective” bulk Hamiltonian,

$$H \simeq H_0 - d_{\vec{r}} (\tilde{S}_{\vec{r}}^x - \tilde{S}_{\vec{r}+\vec{\delta}}^x)$$

- In a linear response approximation this leads to finite expectation value of transverse spin fluctuation.

$$\begin{aligned} \langle \tilde{S}_{\vec{r}}^x \rangle &\sim d_{\vec{r}'} \left(\langle \tilde{S}_{\vec{r}}^x \tilde{S}_{\vec{r}'}^x \rangle - \langle \tilde{S}_{\vec{r}}^x \tilde{S}_{\vec{r}'+\vec{\delta}}^x \rangle \right) \\ &\sim d_{\vec{r}'} \chi_{\vec{r}\vec{r}'} \end{aligned}$$

$$\langle \tilde{S}_{\vec{q}}^x \rangle \sim d_{-\vec{q}} \left(1 - e^{-i\vec{q} \cdot \vec{\delta}} \right) \langle \tilde{S}_{\vec{q}}^x \tilde{S}_{-\vec{q}}^x \rangle \sim d_{-\vec{q}} \chi_{\vec{q}}^d$$

- If we consider the Goldstone modes, $\langle \tilde{S}_{\vec{q}}^x \tilde{S}_{-\vec{q}}^x \rangle = \chi(\vec{q}) \sim \frac{1}{q^2}$, the response is dipolar,

$$\chi_{\vec{q}}^d \sim (\vec{q} \cdot \vec{\delta}) \chi(\vec{q}) \sim \frac{\cos \phi_{\vec{q}}}{q}$$

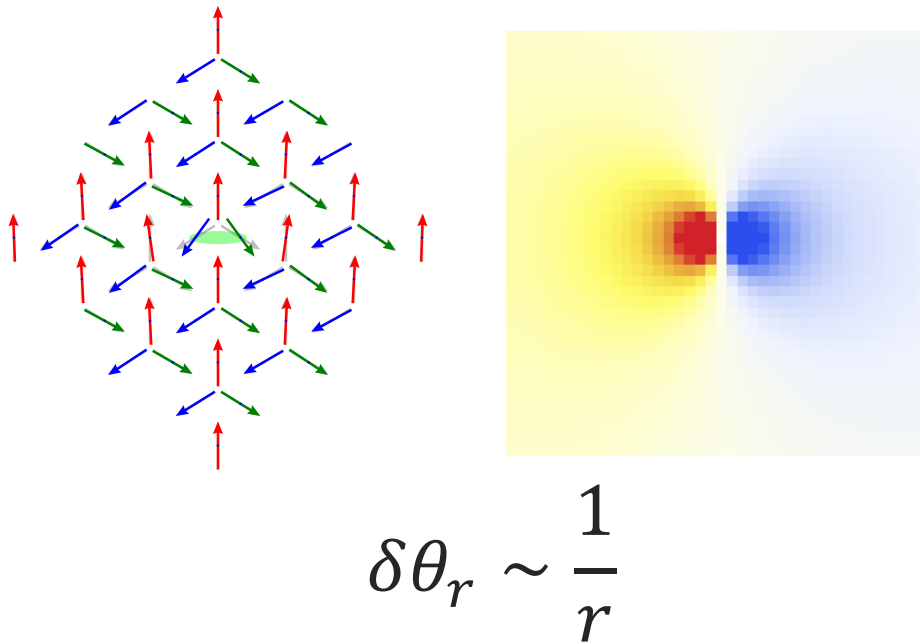
- In 2D, the real space response is also dipolar

$$\chi_{\vec{r}}^d \sim \frac{\cos \theta_{\vec{r}}}{r}$$

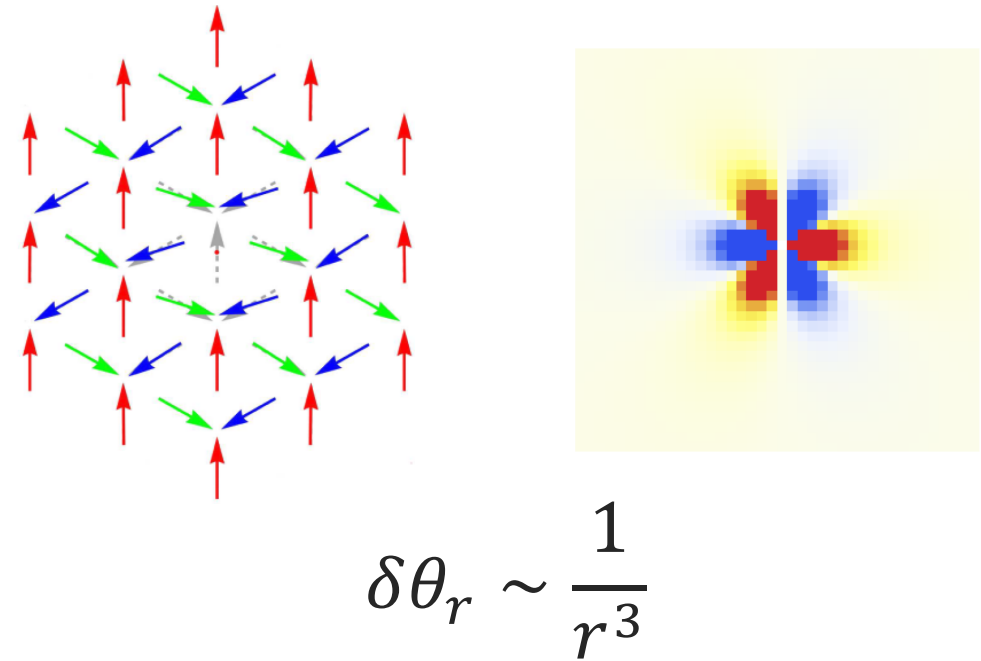
Single bond impurity

Dipolar response in 2D

- Bond impurity: Dipolar response



- Vacancy: Octupolar response



Concentration of bond impurity

Infrared divergence driven by disorder

- Bond impurity: Dipolar response

- For a distribution of random bond impurities,

$$S_{\vec{r}}^{\perp} = \sum_{\vec{r}'} \chi_{\vec{r}\vec{r}'}^d d_{\vec{r}'}$$

- For a disordered system, one expects the impurity magnitudes, $d_{\vec{r}}$ to be uncorrelated.

$$\overline{d_{\vec{r}} d_{\vec{r}'}} = d_0^2 \delta^d(\vec{r} - \vec{r}')$$

This implies that the transverse correlation has a divergent contribution,

$$\begin{aligned} \overline{S_{\vec{r}}^{\perp 2}} &= d_0^2 \sum_{\vec{r}'} \chi_{\vec{r}\vec{r}'}^{d^2} = d_0^2 \int \frac{d^d q}{(2\pi)^d} \chi_{\vec{r}\vec{r}'}^{d^2} \\ &\sim \int d^d q \left(\frac{1}{q}\right)^2 \end{aligned}$$

- The integral diverges for $d \leq 2$.
- Hence the lower critical dimension for bond disordered non-collinear magnets is,

$$d_c = 2$$

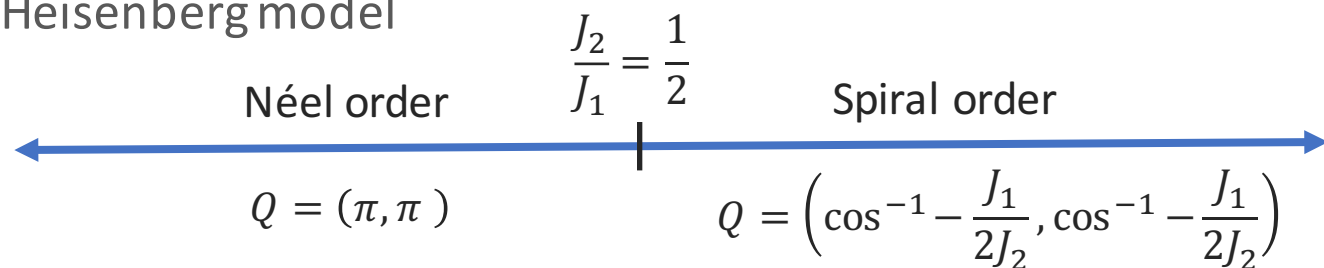
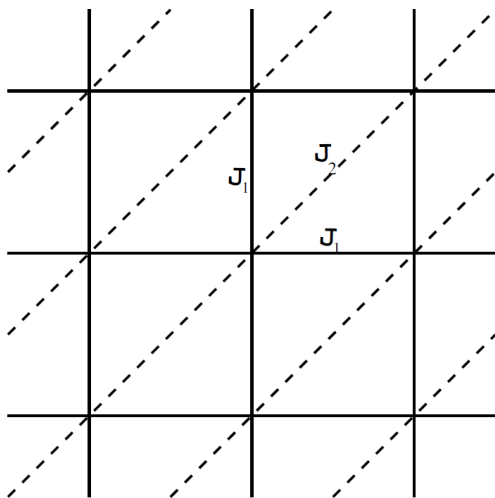
Quenched Disorder: Random Couplings

The Energetics Argument for the Lower Critical Dimension

Incommensurability domains

Incommensurate order in non-collinear magnets

- Our test system: The triangular lattice Heisenberg model



- In the “isotropic” limit, $J_1 = J_2$, the ground state has spiral order with ordering vector,

$$Q = \left(\frac{2\pi}{3}, \frac{2\pi}{3}\right)$$

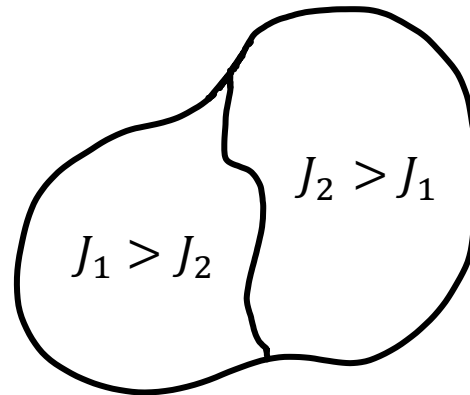
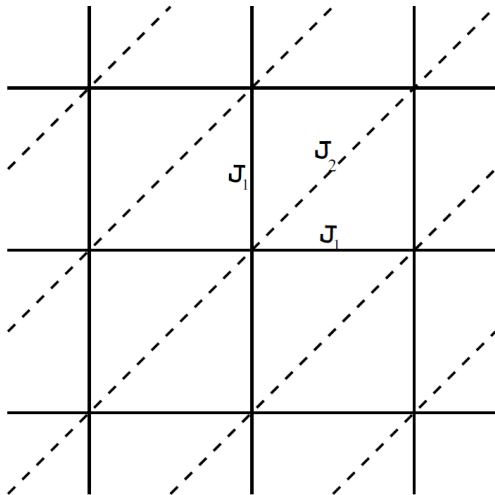
A. Yoshimori, J. Phys. Soc. Jpn. 14, 807 (1959)

Claim: In a system with random couplings, some rare regions emerge with a particular bond anisotropy orientation.

Incommensurability domains

Sharp domain boundaries

- Rare regions with particular bond anisotropy:



- In a lattice there is a discrete number of possible bond anisotropy orientation.
- In a typical sample they define sharp domain walls
- The incommensurate ordering vectors follow the suit

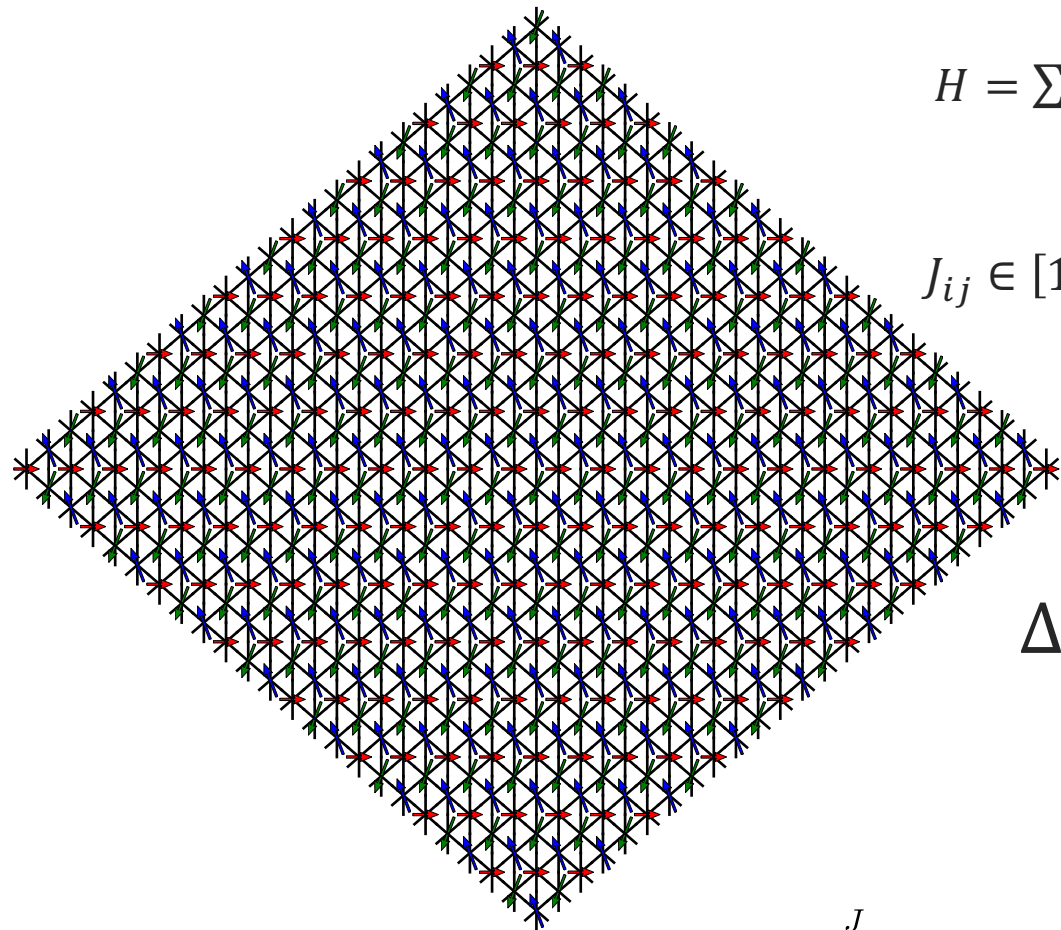
$$Q = \left(\cos^{-1} - \frac{J_1}{2J_2}, \cos^{-1} - \frac{J_1}{2J_2} \right)$$

The surface energy cost for such a domain formation is $\Delta E_{surface} \sim L^{d-1}$, while the bulk energy gain is per site as before, $\Delta E_{bulk} \sim L^{\frac{d}{2}}$. Hence the critical dimension is once again,

$$d_c = 2$$

Incommensurability domains at d=2

Disordered triangular lattice Heisenberg model

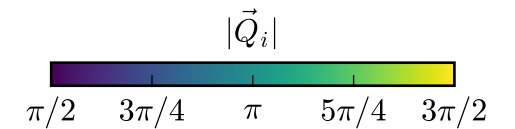
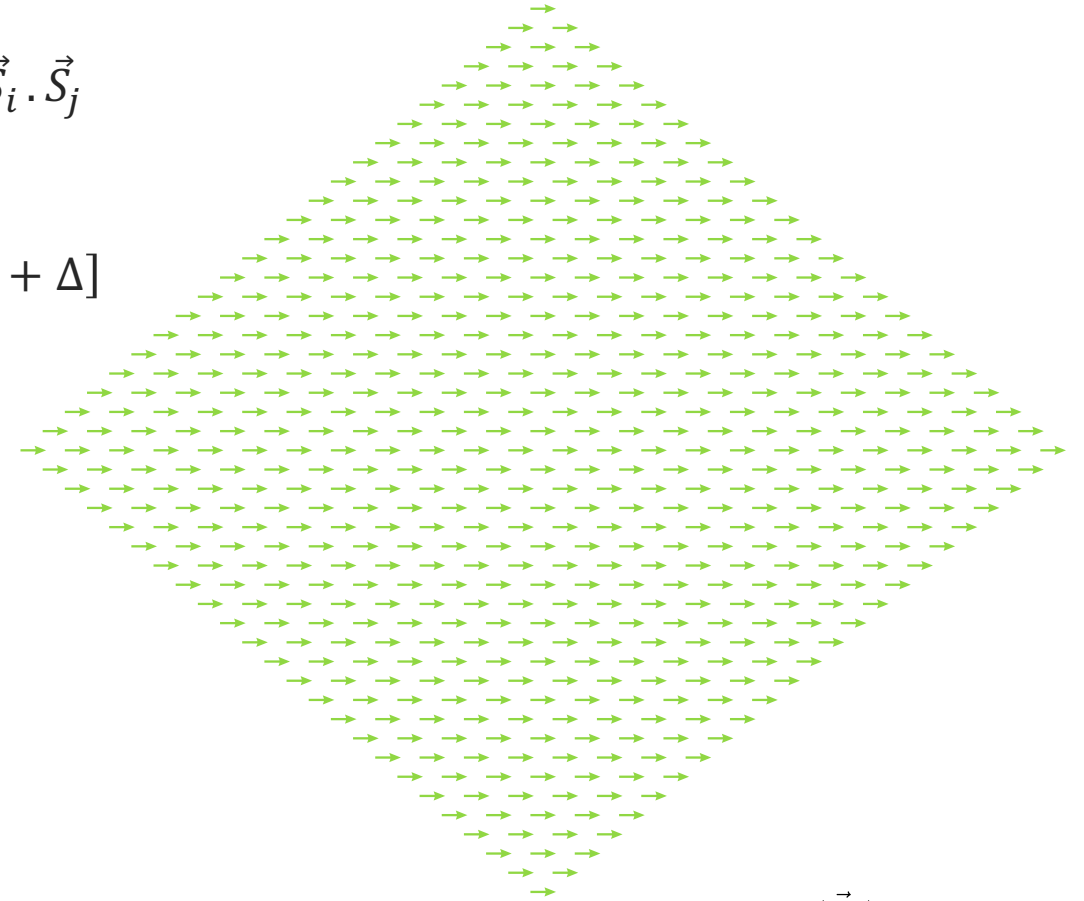
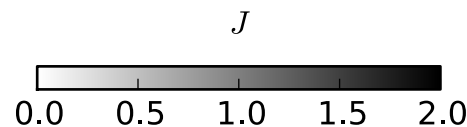


$$H = \sum_{\langle i,j \rangle} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

With

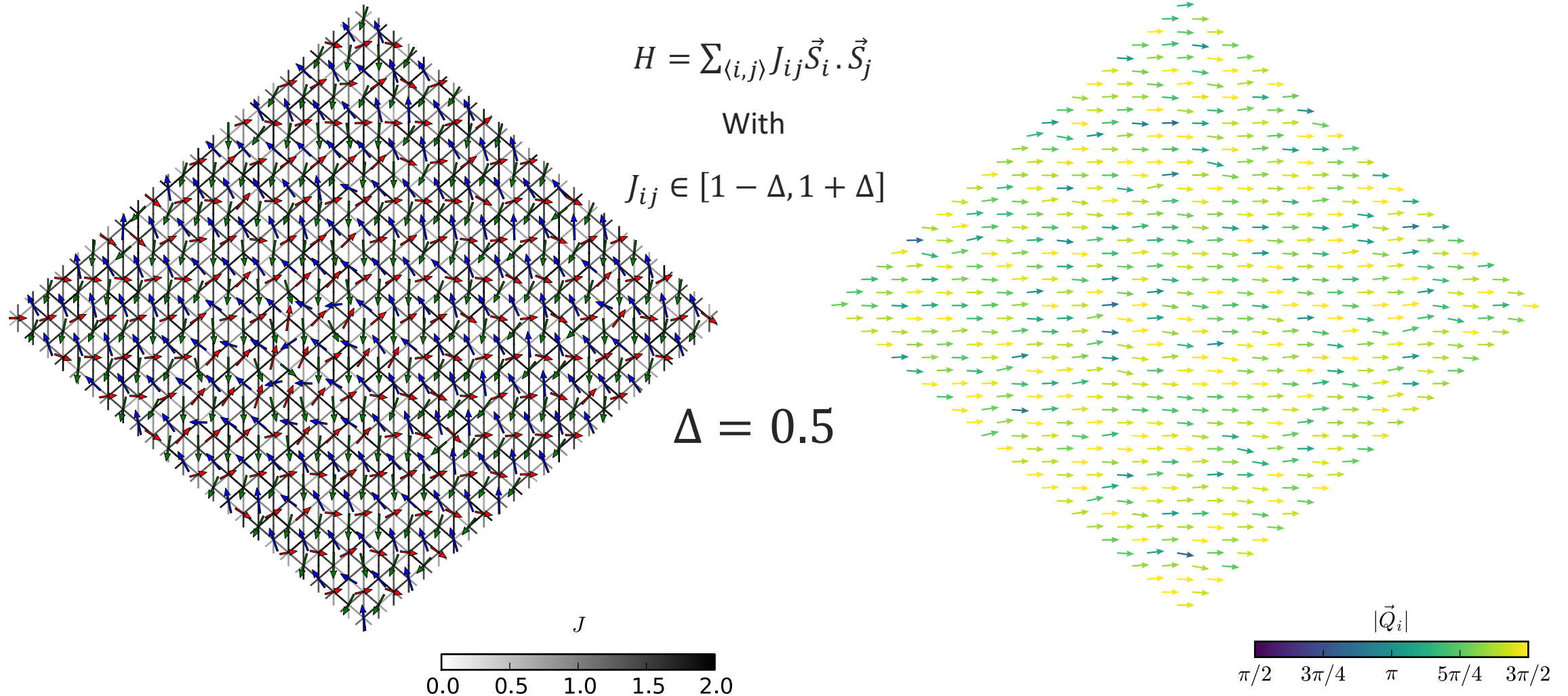
$$J_{ij} \in [1 - \Delta, 1 + \Delta]$$

$$\Delta = 0$$



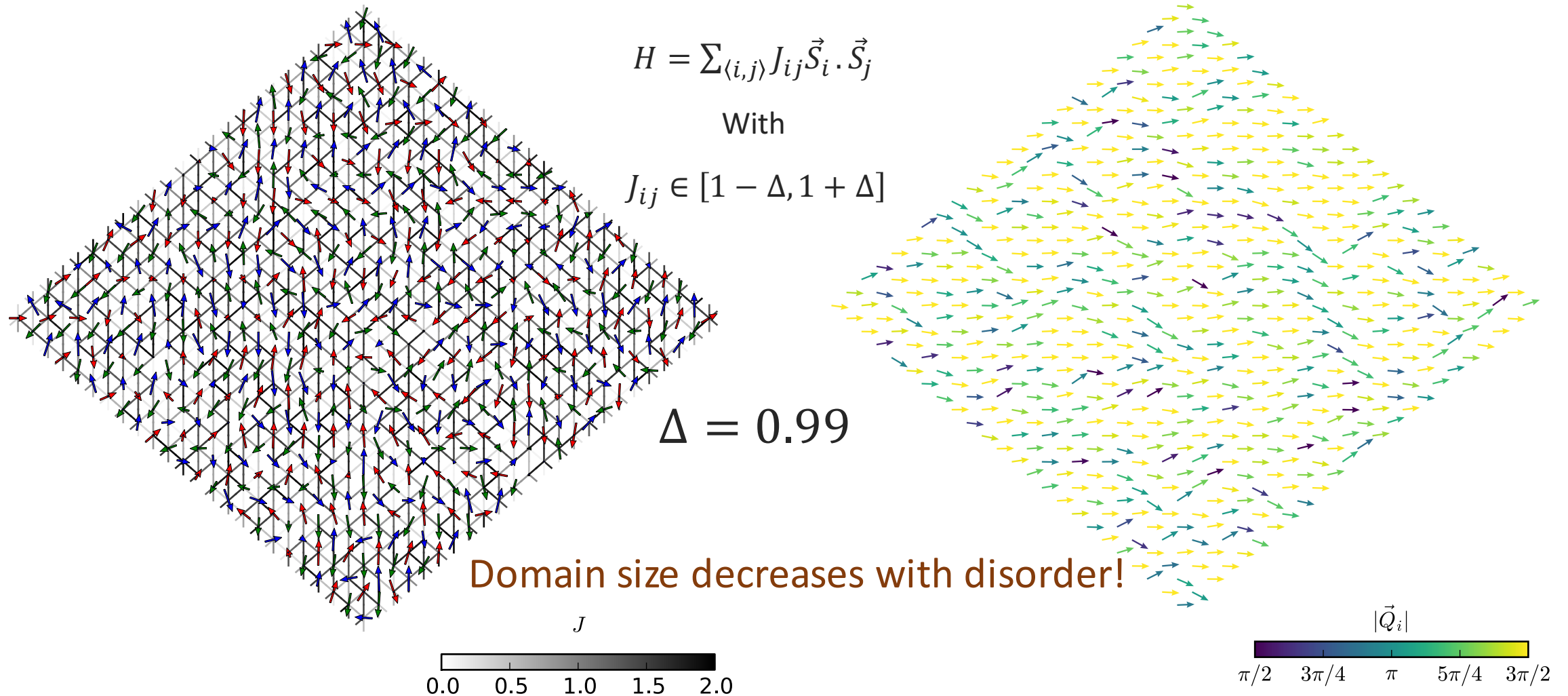
Incommensurability domains at d=2

Disordered triangular lattice Heisenberg model



Incommensurability domains at d=2

Disordered triangular lattice Heisenberg model



Properties at the critical dimension $d_c = 2$

Logarithmic divergence

- Transverse fluctuation diverges logarithmically at $d = 2$

$$\overline{S_{\vec{r}}^{\perp 2}} = d_0^2 \int \frac{d^2 q}{(2\pi)^d} \chi^{d_0^2} \sim d_0^2 \int q dq \left(\frac{1}{q}\right)^2 \sim d_0^2 \log \frac{L}{a}$$

The domain size is determined by the condition, $\overline{S_{\vec{r}}^{\perp 2}} \sim 1$, which defines the domain (correlation) length,

$$\xi_c \sim e^{\frac{1}{d_0^2}}$$

- This leads to the appearance of order for small values of disorder strength.

Properties at the critical dimension $d_c = 2$

Finite temperature

- For finite temperature the system will be thermally disordered.
- However, for small temperature there will be a competition between thermal and quenched disorder.
- At, small T the thermal correlation length is,

$$\xi_c \sim e^{\frac{1}{T}} [1]$$

- So, the crossover temperature is,

$$T^* \sim d_0^2$$

Beyond 2D

Gapped excitations

- If the continuous symmetry of the Hamiltonian is explicitly broken, e.g. an XXZ model, the low energy excitations are gapped, $\chi(q) \sim 1/(q^2 + \lambda)$ and therefore the transverse fluctuation in the presence of random couplings is given by,

$$\overline{S_{\vec{r}}^{\perp 2}} = d_0^2 \int \frac{d^d q}{(2\pi)^d} \chi_{\vec{q}}^2 \sim d_0^2 \int d^d q \left(\frac{q}{q^2 + \lambda} \right)^2$$

- In this case, long range order survives for small disorder with the stability criterion,

$$\overline{S_{\vec{r}}^{\perp 2}} \ll 1$$
$$\Rightarrow \boxed{d_0 \ll \lambda^{2-d}}$$

Beyond 2D

Layered quasi-2D system

- If the coupling between the layers, $J_{\perp}$ is finite then the system has an extra dimension which prevents the divergence and long range order is sustained even in the presence of random couplings. With the bare susceptibility now given by, $\chi(q, q_{\perp}) = \frac{1}{c^2 q^2 + c_{\perp}^2 q^2}$ with $c_{\perp} \propto J_{\perp}$

$$\overline{S_{\vec{r}}^{\perp 2}} = d_0^2 \int \frac{d^d q}{(2\pi)^d} \chi_{\vec{q}}^2 \sim \frac{d_0^2}{J_{\perp}/J} \int d^{d-1} q \frac{1}{q}$$

- The stability criterion, therefore is given by

$$d_0^2 \ll \frac{J_{\perp}}{J}$$

An example

Microscopics and experiments on disordered triangular lattice
antiferromagnet

Disordered triangular lattice Heisenberg model

Disorder distribution

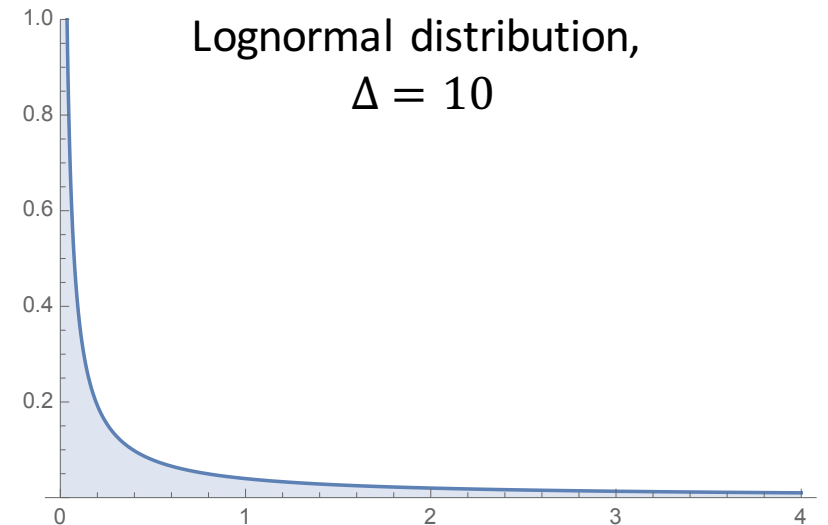
- Disordered $J_1 - J_2$ Heisenberg model

$$H = \sum_{\langle i,j \rangle} J_{ij}^{(1)} \vec{S}_i \cdot \vec{S}_j + \alpha \sum_{\langle\langle i,j \rangle\rangle} J_{ij}^{(2)} \vec{S}_i \cdot \vec{S}_j$$

$$\text{with, } J_{ij}^{(1,2)} \in P(J, \Delta)$$

- From weak to stronger disorder

- Rectangular distribution, $P(J, \Delta) = \frac{1}{2\Delta}, J \in [1 - \Delta, 1 + \Delta]$
- Lognormal distribution, $P(J, \Delta) = \frac{1}{\sqrt{2\pi} (\Delta J)} e^{-\frac{(\log J - 1)^2}{2\Delta^2}}$



Disordered triangular lattice Heisenberg model

Spin-wave theory

- Classical ground state with iterative local energy minimization.
- Spin-wave expansion on top of disordered classical ground state.

$$\tilde{S}_i^z = S - a_i^\dagger a_i$$

$$\tilde{S}_i^+ = \sqrt{2S - a_i^\dagger a_i} a_i$$

- The Hamiltonian becomes,

$$H = H_0 + \sum_i K_i^{(1)} \gamma_i + \sum_{i,j} K_{i,j}^{(2)} \gamma_i \gamma_j + \sum_{i,j,k} K_{i,j,k}^{(3)} \gamma_i \gamma_j \gamma_k + O(\gamma^4) \quad \text{with } \vec{\gamma} = (\vec{a}, \vec{a}^\dagger)$$

- The diagonalization is performed with real space bogoliubov transformation
- The third order terms and the first order terms are combined in a saddle point approximation to give angular fluctuation

Disordered triangular lattice Heisenberg model

Mean-field theory

- Slave fermion representation to capture the spin singlet formation.

$$\vec{S}_i = \frac{1}{2} \sum_{\alpha, \beta} f_{i\alpha}^\dagger \vec{\sigma}_{\alpha, \beta} f_{i\beta}$$

- Decoupling in both singlet and magnetic channel,

$$H = - \sum_{\langle i, j \rangle, \alpha} (\chi_{ij}^{NN} f_{i\alpha}^\dagger f_{j\alpha} + h.c. + NNN) + \frac{1}{2} \sum_{i, \alpha, \beta} \vec{B}_i \cdot f_{i\alpha}^\dagger \vec{\sigma}_{\alpha, \beta} f_{i\beta}$$

- With the mean-field parameters given by,

$$2\chi_{ij}^{NN(NNN)} = (1 - x) \sum_{\alpha} J_{ij} f_{i\alpha}^\dagger f_{j\alpha}, \quad \vec{B}_i = \frac{x}{2} \sum_{j \in NN, NNN, \alpha, \beta} f_{j\alpha}^\dagger \vec{\sigma}_{\alpha, \beta} f_{j\beta}$$

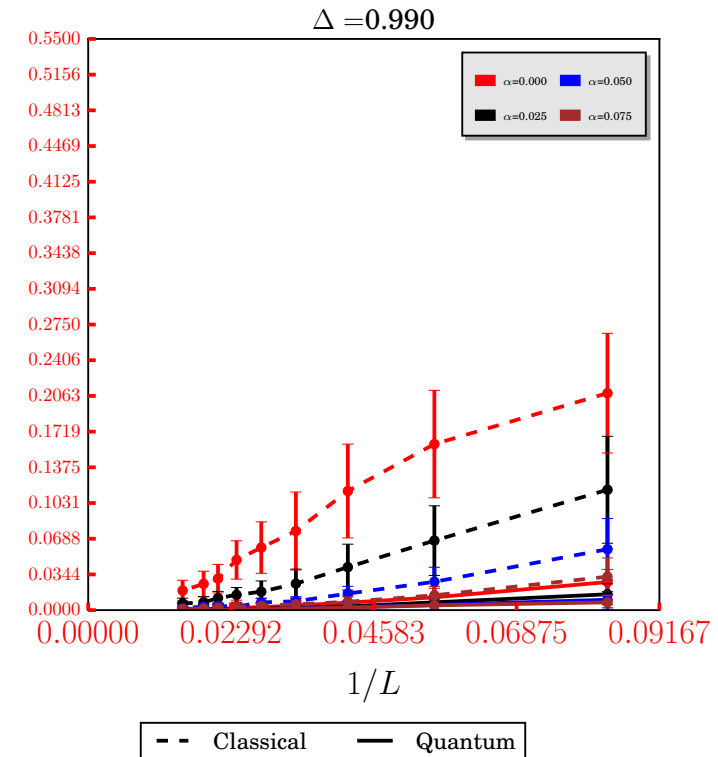
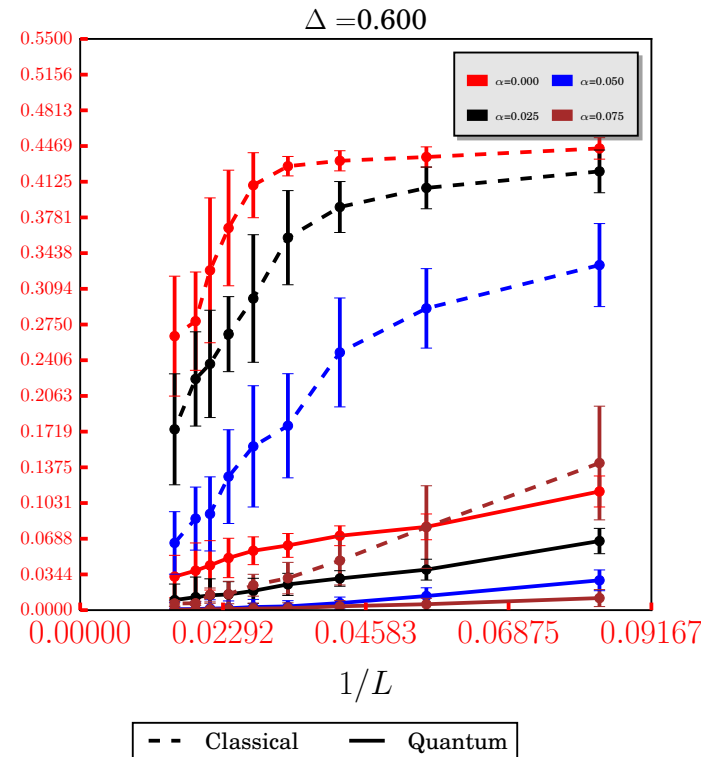
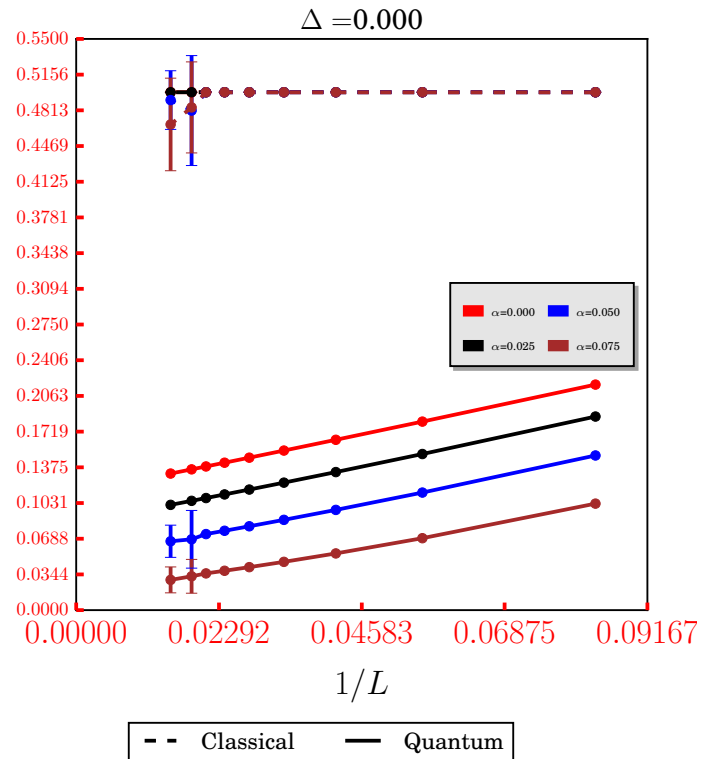
$$\sum_{\alpha} f_{i\alpha}^\dagger f_{i\alpha} = 1$$

$x = \frac{1}{2}$, corresponds to unconstrained Hartree approximation

Disordered triangular lattice Heisenberg model

Amplifying quantum effects

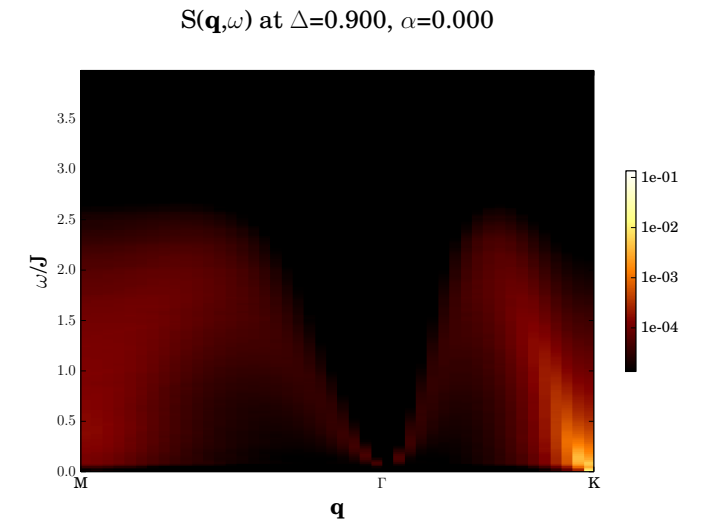
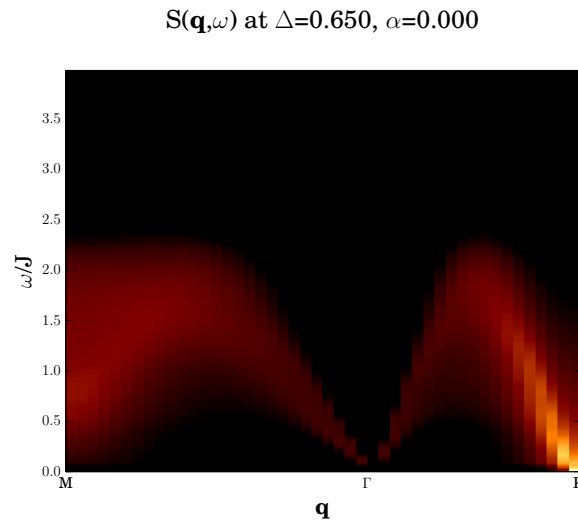
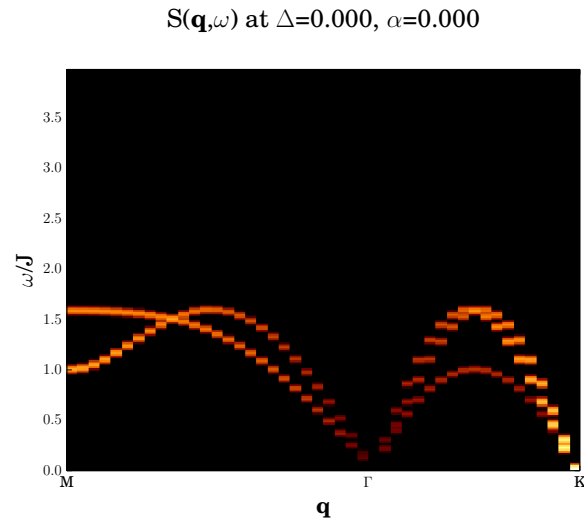
- Structure factor $\chi(Q)/L^2$



Disordered triangular lattice Heisenberg model

Disordered magnon modes

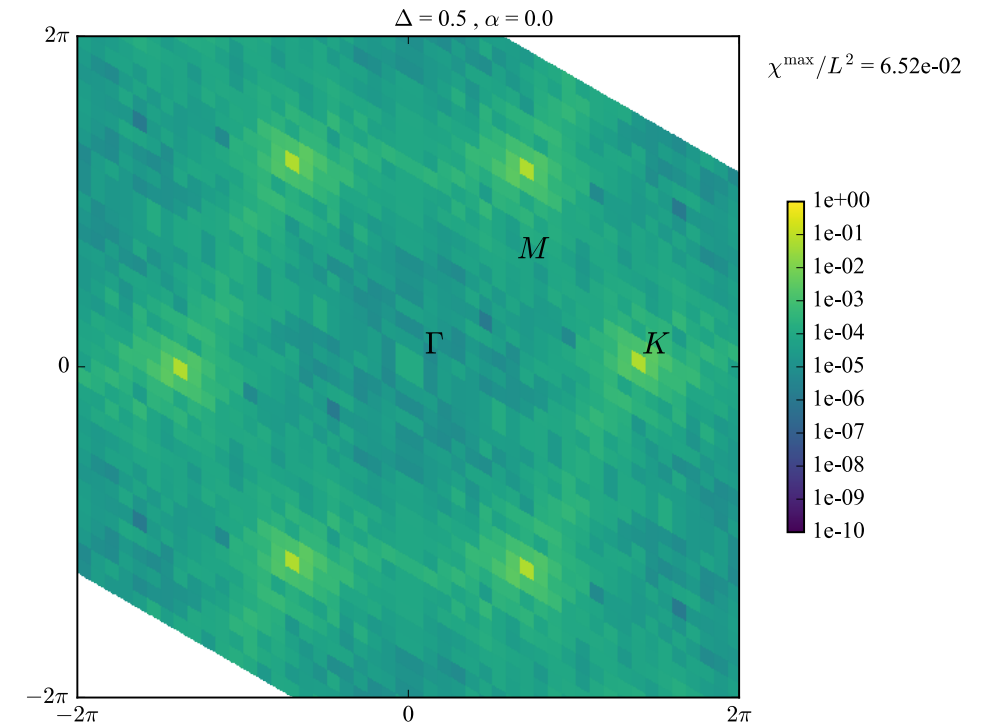
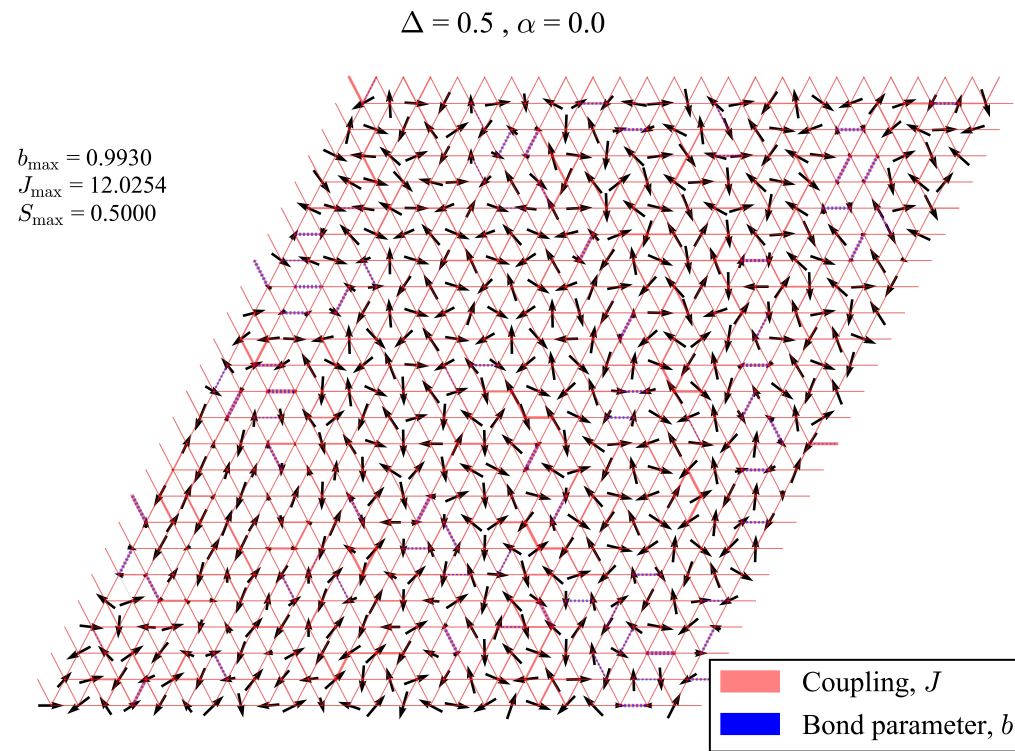
- Dynamic structure factor $\chi(\omega, Q)$



Disordered triangular lattice Heisenberg model

Composite glass

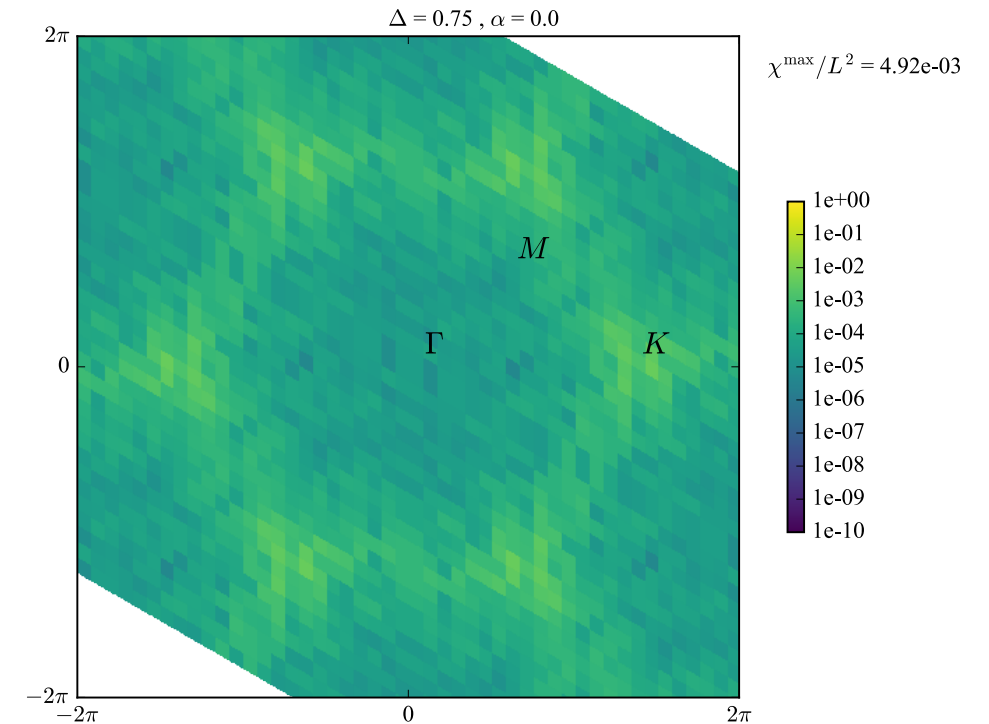
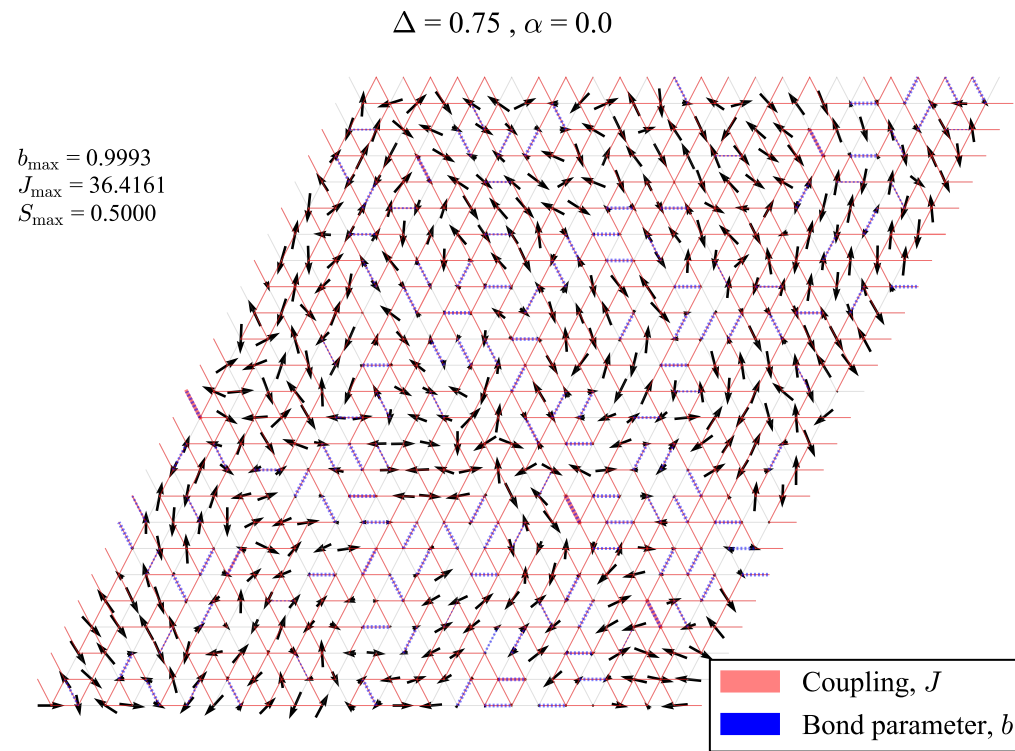
- Valence bonds at higher disorder



Disordered triangular lattice Heisenberg model

Composite glass

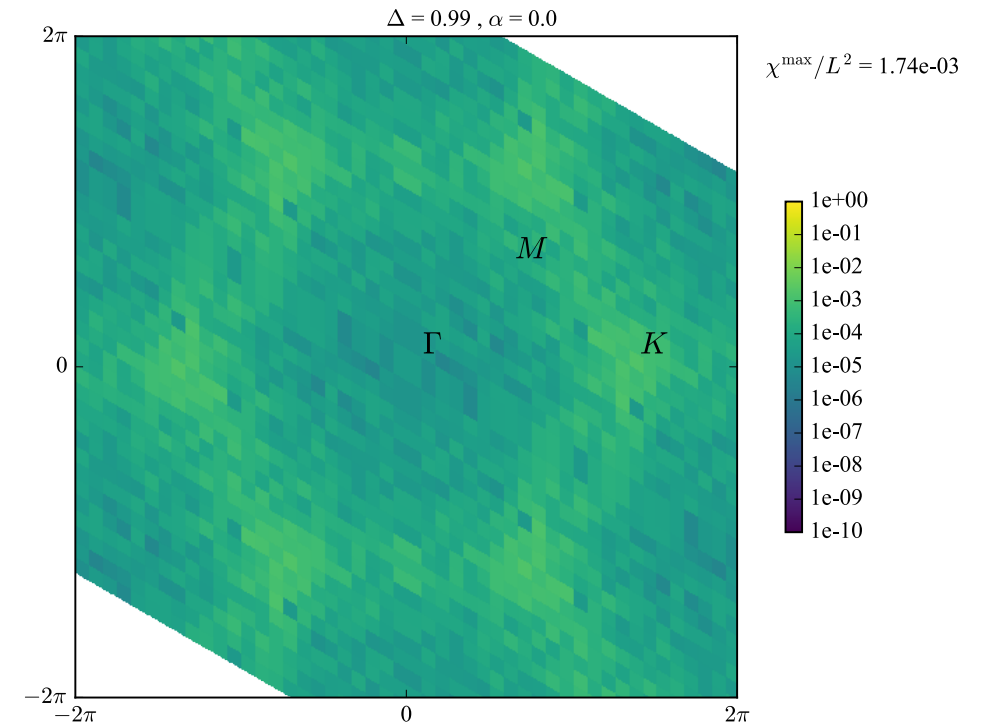
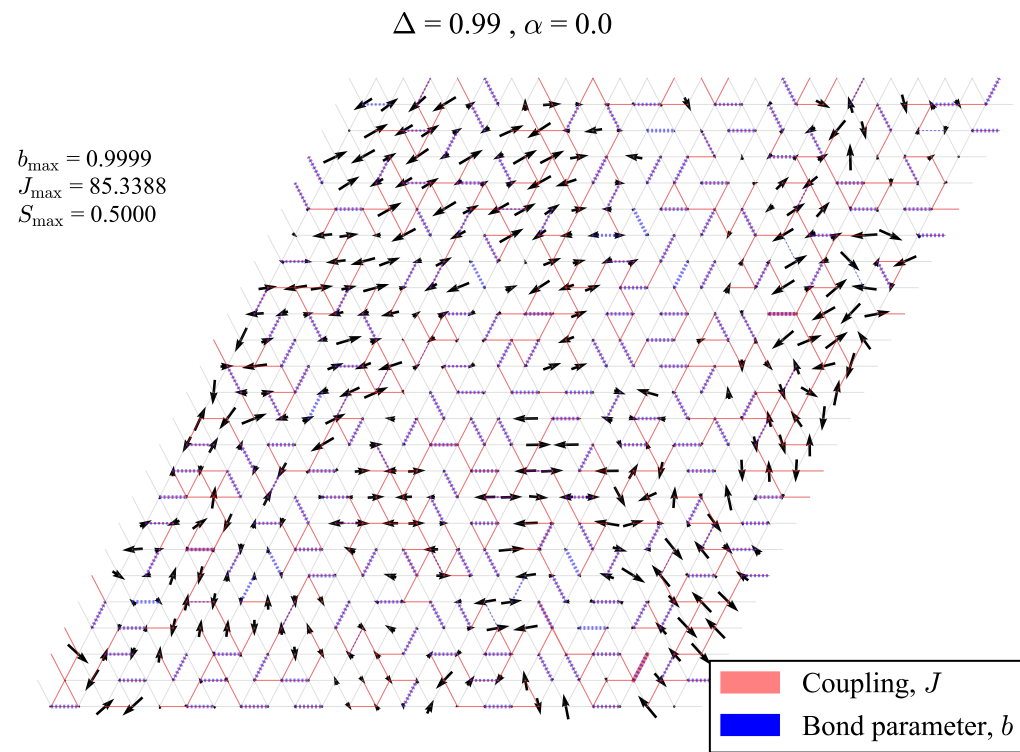
- Valence bonds at higher disorder



Disordered triangular lattice Heisenberg model

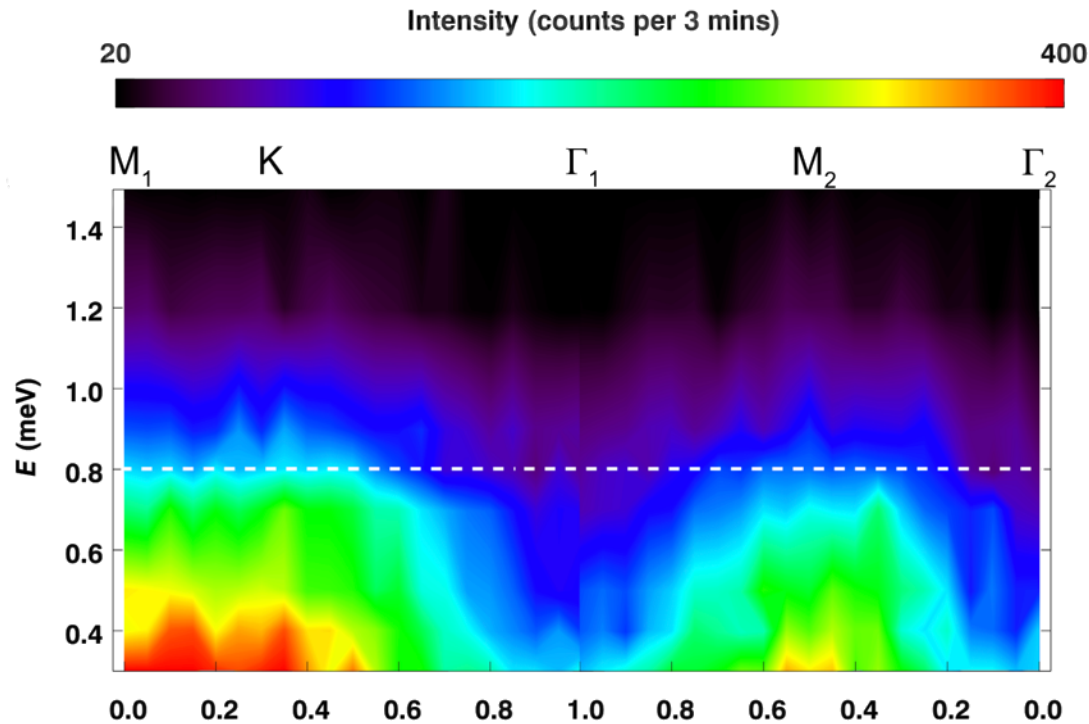
Composite glass

- Valence bonds at higher disorder



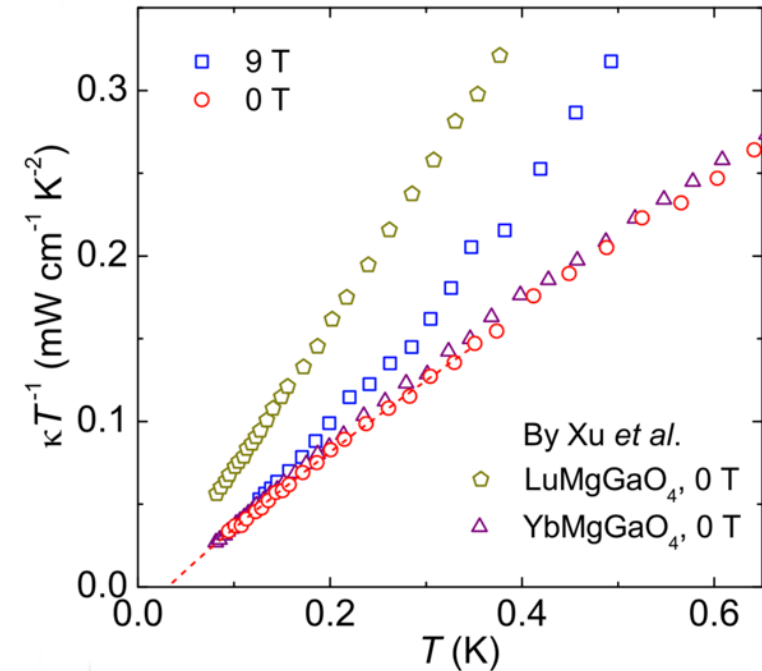
Experiments on triangular lattice

Spin-glass signature in $YbZnGaO_4$



Magnon modes

Ma et. al, arxiv:1709.00256 (2017)

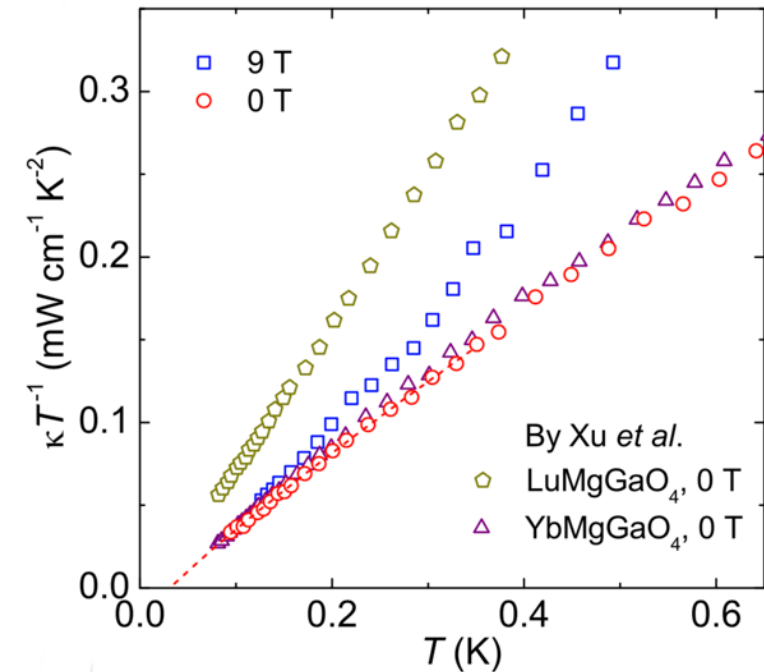


Thermal conductivity, $\frac{\kappa}{T} = \frac{\kappa_0}{T} + n T^{\beta-1}$

Experiments on triangular lattice

Spin-glass signature in $YbZnGaO_4$

- Very small non-phonon contribution $\frac{\kappa_0}{T}$
- 1000 times smaller than QSL candidate $EtMe_3Sb[Pd(dmit)_2]$
- It is postulated that disorder induced spin Glass signature might explain this behavior.



Thermal conductivity, $\frac{\kappa}{T} = \frac{\kappa_0}{T} + n T^{\beta-1}$

Summary

- Random bond disorder destroys order in non-collinear magnets below or at the critical dimension 2.
- However, relatively large domain sizes at the critical dimension creates a semblance of an ordered phase.
- Quantum fluctuation makes the effect of disorder more pronounced.

Thank you!

